

The Reconstruction Will Not Be Supervised

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Introduction

The prehistoric human past is accessible through three complementary windows: **archaeology**, **population genetics**, and the **Comparative Method**—a set of techniques developed by historical linguists over the last 150 years to reconstruct unattested ancestral languages from their modern descendants. This handout summarizes three studies toward automating the Comparative Method with machine learning.



Figure 1: Scientists were able to understand the Austronesian expansion using the Comparative Method, population genetics, and archaeology.

Theory

Cognate Sets and Protoforms

Words in related languages that descend from a single ancestral word are **COGNATES**. A **COGNATE SET** is the collection of cognates across all languages in a family that share a single origin. The inferred ancestral word is called a **PROTOFORM**.

Gloss	English	Latin	Greek	Sanskrit	Avestan
‘two’	two	duo	dúō	dvā(u)	dva
‘tooth’	tooth	dens	odón	dán	dantan
‘tree’	tree	—	doru	dāru	dāru
‘ten’	ten	decem	déka	dáśa	dasa

Table 1: Selected Indo-European cognate sets. The systematic initial-consonant correspondences (e.g., t:d:d across Germanic, Italic, Greek, and Indo-Iranian) motivate positing a common ancestral form.

Matching these patterns across the three cognate sets for ‘daughter’, ‘tooth’, and ‘three’ yields reconstructed Proto-Indo-European protoforms such as **d^hugh₂tēr*, **h₃dónts*, and **tréyes*.

	English		Greek		Sanskrit		Proto-Indo-European
	d daughter	t ^h	t ^h ugátēr	d ^h	d ^h úhitṛ	*d ^h	*d ^h ugh ₂ tēr
	t tooth	d	odón	d	dántas	*d	*h ₃ dónts
	θ three	t	treĩs	t	tráyas	*t	*tréyes

Table 2: Three correspondence sets for Indo-European ‘daughter’, ‘tooth’ and ‘three’

The Goal: Learning $\pi(\cdot)$ and $\rho(\cdot)$

The Comparative Method requires jointly learning two functions:

$$\pi(\mathcal{K}_i) = p_i \quad (\text{encoder: cognate set} \rightarrow \text{protoform})$$

$$\rho(p_i) \approx \mathcal{K}_i \quad (\text{decoder: protoform} \rightarrow \text{cognate set})$$

such that $\rho(\pi(\mathcal{K}_i)) \approx \mathcal{K}_i$ and $\pi(\rho(p_i)) \approx p_i$. Reconstruction is thus structurally analogous to an *autoencoder* in which π is the encoder, ρ is the decoder, and protoforms are the latent representation.

An optimal reconstruction satisfies $H(\mathcal{K} \mid P) \approx 0$ and $H(P \mid \mathcal{K}) \approx 0$: cognate sets and protoforms are lossless encodings of each other. Reconstruction is a generative modeling problem subject to (1) **unsupervised compression** and (2) **interpretability via symbolic rules**.

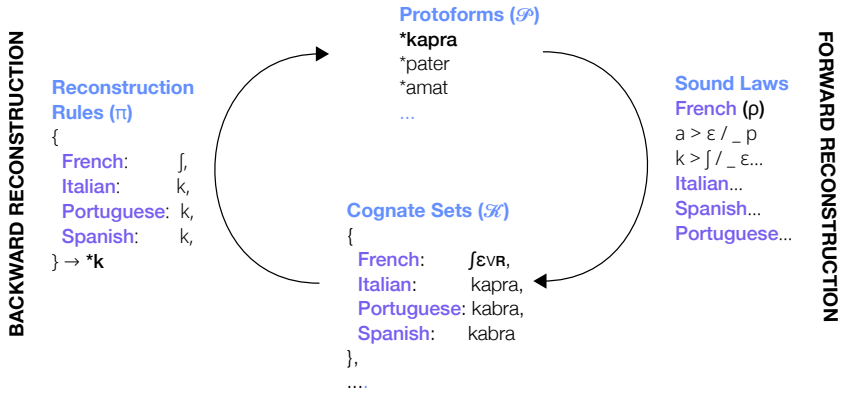


Figure 2: A formalization of the comparative method.

The Regularity Hypothesis and Rewrite Trees

The **REGULARITY HYPOTHESIS** (Neogrammarian Hypothesis) constrains the space of valid ρ functions: sound change proceeds mechanically and deterministically. Therefore $\rho(\cdot)$ must be expressible as a minimal *rewrite tree*—a tree of sound laws, each of the form $L\alpha R \rightarrow L\beta R$:

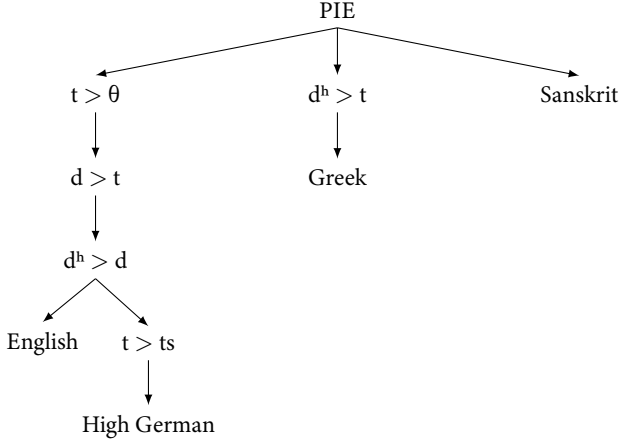


Figure 3: A portion of the Indo-European rewrite tree $\rho(\cdot)$. Each internal node is a sound law; leaf nodes are descendant languages. The path from `PIE` to High German traverses four sound laws in sequence.

The tree in Figure illustrates the mutual dependence of π and ρ : the protoforms posited by π must be similar enough to the observed reflexes that ρ can be stated as a *minimal* program. This requires solving both simultaneously.

Three Studies

The three studies presented here address the learning of $\pi(\cdot)$ and $\rho(\cdot)$ from complementary perspectives:

1. **Semisupervised joint learning** of π and ρ (Study 1)
2. **Zero-shot sound law induction** by framing ρ as a programming-by-example problem (Study 2)
3. **Benchmarking LLM reasoning** on ρ -induction as a pure search task (Study 3)

Study 1: Semisupervised Neural Protoform Reconstruction

Motivation

Early neural approaches to learning $\pi(\cdot)$ were fully supervised sequence-to-sequence models [Meloni et al., 2021, Chang et al., 2022]. The bottleneck is labeled data: annotating cognate sets with gold protoforms requires substantial expert effort, and far more unannotated cognate sets exist than labeled ones. This study introduces the `D2P2D` architecture, which exploits unlabeled cognate sets in a semisupervised framework.

Liang Lu, Peirong Xie, and David R. Mortensen.
ACL 2025 (Best Paper Award).

Dataset	Sets	Ancestor
WikiHan [Chang et al., 2022]	5,165	Middle Chinese
Hóu	804	Middle Chinese
Rom-phon [Meloni et al., 2021]	8,703	Latin
Rom-orth [Meloni et al., 2021]	8,631	Latin

Table 3: Datasets for Study 1. Each dataset includes cognate sets across 5–39 daughter languages. Labels (gold protoforms) are available for all sets; semisupervised experiments withhold 70–95% of labels.

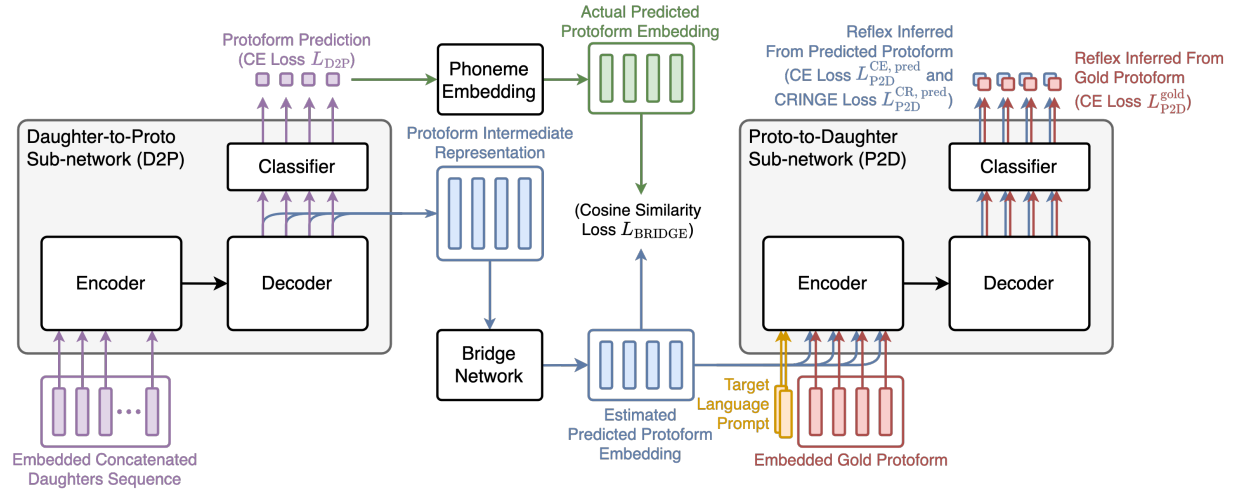


Figure 4: The D2P2D architecture. The D2P sub-model encodes cognate sets as protoforms; the *bridge network* maps D2P decoder output to estimated protoform embeddings; the P2D sub-model decodes those embeddings back into reflexes. When labeled protoforms are available, gold embeddings supervise the bridge via a cosine similarity loss.

Datasets

Architecture

The overall training objective combines five losses:

$$L = \alpha_1 L_{D2P} + \alpha_2 L_{P2D}^{CE} + \alpha_3 L_{P2D}^{CR} + \alpha_4 L_{P2D}^{gold} + \alpha_5 L_{bridge}$$

where the constants $\alpha_{1...5}$ weight each term. For unlabeled cognate sets, only the first three terms contribute; the model still learns from the structural constraint that $\rho(\pi(\mathcal{K}_i)) \approx \mathcal{K}_i$.

Baselines

supv Naïve supervised seq2seq model (previous state of the art).

bst Bootstrapping: add the model’s most confident unlabeled predictions to the training set.

Π M Π -Model: minimize consistency between two stochastically augmented copies of each training example.

dpd The D2P2D architecture described above.

dpd- Π M-bst DPD combined with Π -Model and Bootstrapping.

Results

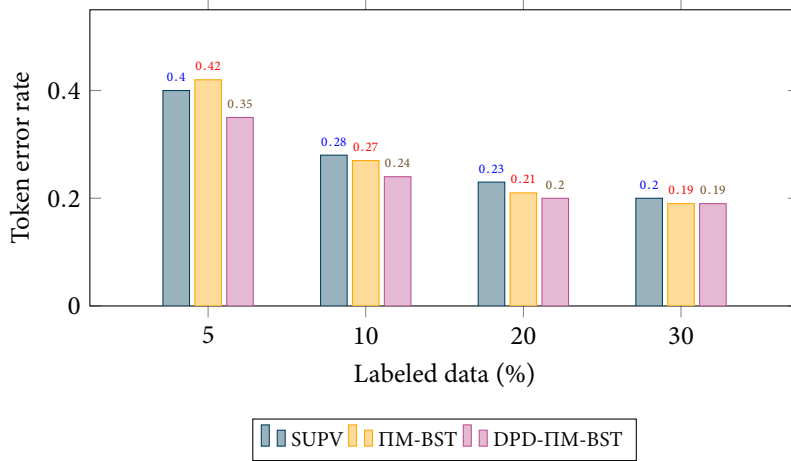


Figure 5: Token error rate on WikiHan (transformer sub-model; lower is better). DPD- Π M-BST (purple) outperforms SUPV (blue) and Π M-BST (gold) at all labeled-data conditions; gains are largest in the low-label regime.

DPD variants achieve the best average accuracy across all data settings, language families, and sub-model architectures; the DPD- Π M-BST combination is the overall winner. The result confirms that the mutual constraint between π and ρ provides a strong unsupervised learning signal.

Study 2: Sound Law Induction via Programming by Example

Motivation

Study 1 treats $\pi(\cdot)$ as a black-box sequence model. Practitioners, however, need $\rho(\cdot)$ in interpretable form—as a sequence of rewrite rules. Crucially, formulating $\rho(\cdot)$ is *precisely what linguists do* when they write sound laws. Each sound law $L\alpha R \rightarrow L\beta R$ corresponds directly to a Python string replacement:

```
s = re.sub(r"(?<=L)a(?=R)", "b", s)
```

Sound law induction therefore reduces to **PROGRAMMING BY EXAMPLE (PBE)**: given input-output pairs of protoforms and their reflexes, generate the Python program that implements the sound laws.

Structure vs. Substance

Fine-tuning a code language model for PBE requires a training distribution. Two orthogonal dimensions vary:

Structure Programs are structurally similar to sound laws ($\alpha \rightarrow \beta$ schema; inputs drawn randomly from Σ^+) but otherwise random.

Substance Programs are drawn from collections of attested sound laws (*Index Diachronica*); inputs resemble actual or nonce protoforms from real languages.

The working hypothesis is that **substance** > **structure**: inductive biases toward linguistically plausible programs and language-like inputs should improve generalization to real language data.

Condition	Description
RP-RI	Random programs; random string inputs
RP-LI	Random programs; LLM-generated nonce protoform inputs
RP-PI	Random programs; actual proto-language inputs
IDP-PI	<i>Index Diachronica</i> programs; proto-language inputs

Atharva Naik, Rebecca Byrnes, Anna Cai, Calvin Chang, Clayton Marr, Aravind Mysore, Nate Robinson, Hong Sng, and Carolyn Rosé.

Table 4: Finetuning conditions, ordered by increasing substance. RP-LI represents the optimum: structurally valid programs paired with language-like (but not task-specific) inputs.

Test Data

Test sets were constructed from two language families:

- Proto-Polynesian $\rightarrow$ Hawaiian, Niue, Samoan, Tongan
- Proto-Tangkhalic $\rightarrow$ Huishu

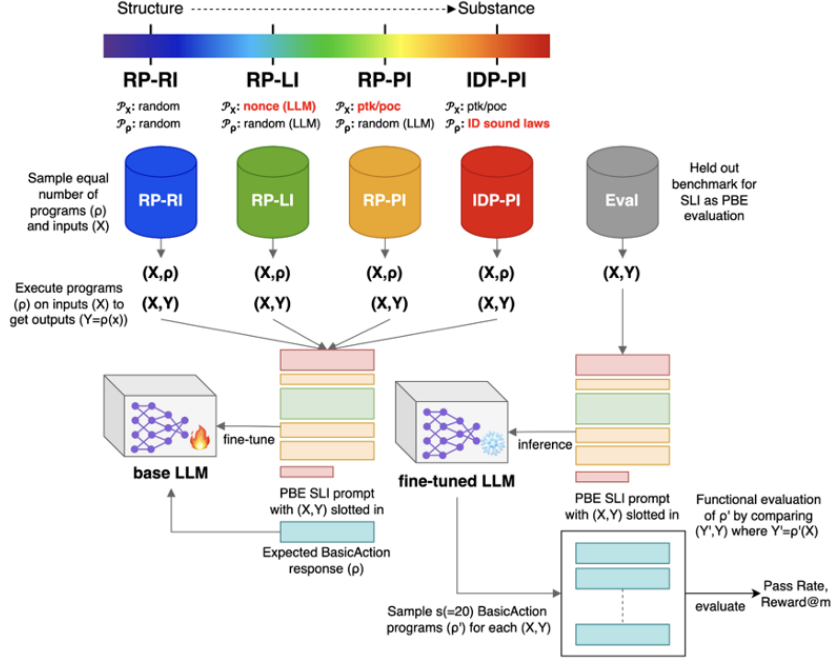


Figure 6: Overview of the PBE pipeline. A code language model is prompted with input-output word pairs and must generate a Python program (cascade of re. sub calls) implementing the sound laws.

Reference programs were handcrafted with assistance from the computational forward-reconstruction tool DiaSim [Marr and Mortensen, 2022]. Single-law datasets were constructed by iterating over sound laws in each reference program, retaining all changed words at each step as input-output pairs.

Results

Model	Haw.	Niue	Sam.	Tong.	Hui.	Avg
Codestral-22B	54.9	100	55.6	50.0	50.4	62.2
PySLICoder (codestral)	52.9	100	44.5	37.5	60.5	59.1
PySLICoder (gpt-4o)	82.4	100	68.4	37.5	58.1	68.4
GPT-4o	90.2	100	72.2	75.0	72.9	82.1

The best fine-tuned model (PySLICoder-RP-LI, trained with GPT-4o-generated inputs) achieves 68.4% average pass rate, approaching GPT-4o (82.1%). The best finetuning condition is RP-LI—random programs with LLM-generated nonce inputs. Models fine-tuned on *Index Diachronica* programs (IDP-PI) do not outperform RP models, possibly due to overfitting to that compilation’s distribution.

Table 5: Pass rate (%) on single sound-law induction as PBE. PySLICoder variants are fine-tuned on the RP-LI condition. Haw. = Hawaiian; Sam. = Samoan; Tong. = Tongan; Hui. = Huishu (Proto-Tangkhulic descendant).

Study 3: PBE Bench—A Reasoning Benchmark for LLMs

Forward Reconstruction as a Pure Search Problem

Sound law induction as PBE can also serve as a **pure search benchmark** for LLMs. Unlike most reasoning benchmarks (which conflate search with domain knowledge), forward reconstruction requires only search: given input-output pairs from an unknown program over strings, find the program. No linguistic knowledge is required.

Rule Ordering Relations

Multi-rule cascades interact. All pairwise ordering relationships between rules p_i and p_j can be classified as follows:

Feeding (F) p_i creates substrings that enable p_j to apply.

Bleeding (B) p_i removes substrings required by p_j .

Counter-Feeding (CF) p_i could feed p_j , but p_j precedes p_i in the cascade.

Counter-Bleeding (CB) p_i could bleed p_j , but p_j precedes p_i in the cascade.

PBE Bench-Lite balances instances across these four relation types to measure their relative contribution to task difficulty.

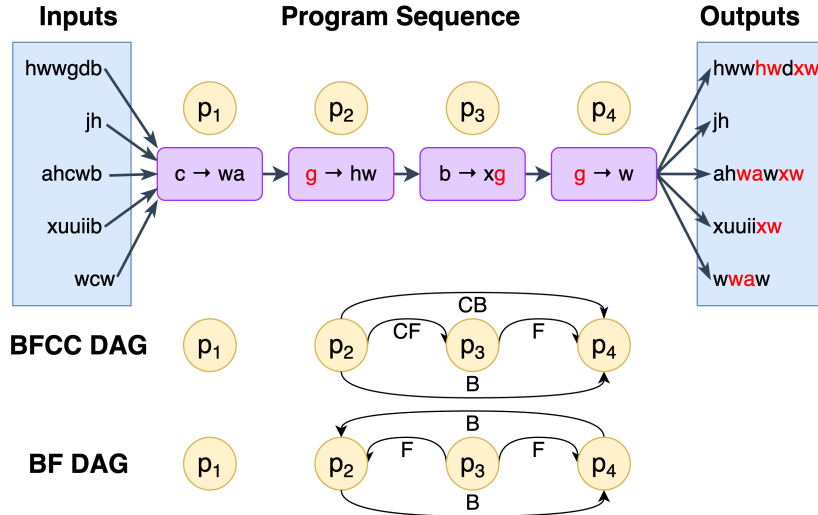


Figure 7: A PBE Bench data instance. The model receives input-output word pairs from an unknown cascade of rewrite rules and must produce a Python program implementing those rules. Labels B, F, CB, and CF (not visible to the model) indicate the ordering relation between successive rules.

Results on PBE Bench-Lite

Scaling Analysis

Model	Pass@1
Qwen3-30B-A3B	0.29
Claude 4 Sonnet	0.30
Claude 3.7 (Thinking)	0.33
Claude 4 (Thinking)	0.36
QwQ-32B	0.36
o4-mini	0.36
Qwen3-32B (CoT)	0.39
gpt-oss-20B	0.41
Gemini 2.5 Flash	0.58
gpt-oss-120B	0.63
GPT-5	0.72

Table 6: Pass@1 on PBE Bench-Lite (first-code-block evaluation), ranked by increasing performance. Most models fall in the 0.29–0.41 range; GPT-5 and gpt-oss-120B are clear leaders.

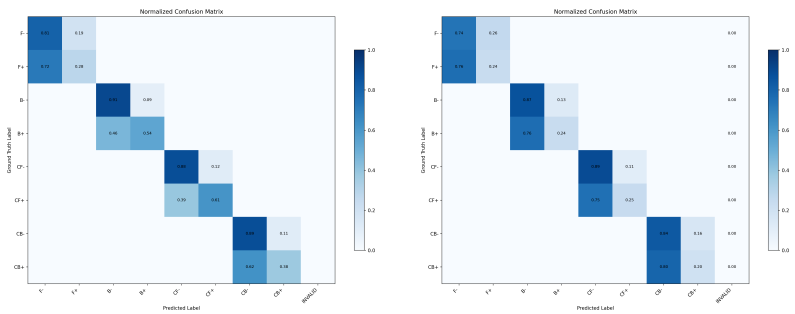


Figure 8: Confusion matrices for rule-ordering relations across PBE Bench-Lite successes (left) and failures (right). Bleeding (B) and counter-bleeding (CB) relations are associated with proportionally more failures, reflecting their greater logical complexity.

Conclusions

A reconstruction is **correct** if and only if:

- $\pi(\rho(p_i)) \approx p_i$ for all protoforms $p_i \in \Sigma^*$
- $\rho(\pi(K_i)) \approx K_i$ for all cognate sets $K_i \in \mathcal{K}$
- $\rho(\cdot)$ is a *minimal* program of sound laws

The three studies presented here make complementary progress toward satisfying these criteria. Study 1 demonstrates that the mutual constraint between π and ρ enables effective semisupervised learning. Study 2 shows that $\rho(\cdot)$ can be recovered as interpretable Python code via PBE fine-tuning, with language-like training inputs being more important than substantively realistic sound laws. Study 3 establishes a rigorous search benchmark, revealing that state-of-the-art LLMs handle cascades of roughly 20 rules, whereas actual historical reconstruction may involve 300 or more [Marr and Mortensen, 2022]. Bridging this gap is the central challenge for the next generation of methods.

Reconstruction Agents

Given that we can formalize the Comparative Method, use LLMs to implement ρ , and use neural models to implement π , why not frame the entire enterprise as an iterative refinement problem and employ coding agents to search for approximate solutions?

We can state the reward function for this task as follows:

$$\mathcal{Q}(\mathcal{K}, \pi, \rho) = \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m [\rho(\pi(K_i)) = K_i] [k_i^{(j)} \in V_{\ell^{(j)}}] - \alpha|\pi| - \beta|\rho|, \quad (1)$$

$$\langle \mathcal{K}, \pi, \rho \rangle = \arg \max_{\mathcal{K}, \pi, \rho} \mathcal{Q}(\mathcal{K}, \pi, \rho). \quad (2)$$

where $\mathcal{K} = \{K_1, K_2, \dots, K_n\}$ is the collection of cognate sets, $k_i^{(j)}$ is the reflex of K_i in language $\ell^{(j)}$, $V_{\ell^{(j)}}$ is the vocabulary of the j th language, and $|\pi|$ and $|\rho|$ denote the complexity of π and ρ as measured by the number of expressions.

References

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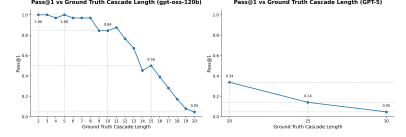


Figure 9: Pass@1 vs. cascade length for gpt-oss-120B (left) and GPT-5 (right). Performance degrades sharply beyond approximately 20 rules. Historical sound-change cascades may contain 300 or more rules [Marr and Mortensen, 2022].

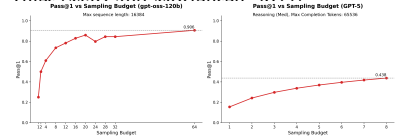


Figure 10: Pass@k vs. sampling budget (cascade length 10, max sequence length 16K/65K tokens) for gpt-oss-120B (left) and GPT-5 (right). Increasing the sampling budget yields substantial gains for both models.

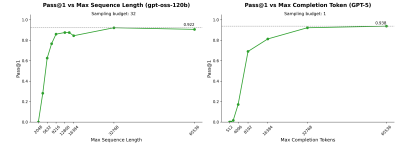


Figure 11: Pass@1 vs. maximum sequence length (cascade length 10, sampling budget 1) for gpt-oss-120B (left) and GPT-5 (right). Longer context windows consistently improve performance; both models benefit substantially from contexts of at least 16K tokens.

Initialized					Round 1					Round 2				
'sharpen' 'weave' 'bone' 'ten'					'sharpen' 'weave' 'bone' 'ten'					'sharpen' 'weave' 'bone' 'ten'				
Tusom	te	to	ru	he	Tusom	te	to	ru	he	Tusom	te	to	ru	he
Kachai	ðu	ðøk	rø	ru	Kachai	ðu	ðøk	rø	ru	Kachai	ðu	ðøk	rø	ru
Huishu	re	ro?	ruk	re	Huishu	re	ro?	ruk	re	Huishu	re	ro?	ruk	re
Ukhrul	ra	rak	ru	ra	Ukhrul	ra	rak	ru	ra	Ukhrul	ra	rak	ru	ra
π:	*r	*r	*r	*r	π:	*ð	*ð	*r	*h	π:	*ð	*ð	*r	*r
ρ:	Kachai: r > ð				ρ:	Tusom: ð > t Kachai: h > r				ρ:	Tusom: ð > t,			
	Tusom: r > t, r > h					Huishu: h > r Ukhrul: h > r					r > h / _ e			
	Errors: 4					Errors: 0					Errors: 0			
PRED	'sharpen'	'weave'	'bone'	'ten'	PRED	'sharpen'	'weave'	'bone'	'ten'	PRED	'sharpen'	'weave'	'bone'	'ten'
Tusom	te	to	ru	he	Tusom	te	to	ru	he	Tusom	te	to	ru	he
Kachai	ðu	ðøk	rø	ru	Kachai	ðu	ðøk	rø	ru	Kachai	ðu	ðøk	rø	ru
Huishu	re	ro?	ruk	re	Huishu	re	ro?	ruk	re	Huishu	re	ro?	ruk	re
Ukhrul	ra	rak	ru	ra	Ukhrul	ra	rak	ru	ra	Ukhrul	ra	rak	ru	ra

Figure 12: Stepping through an iterative refinement of a reconstruction.

Heng Ji, Sadao Kurohashi, Patrizia Paggio, Nianwen Xue, Seokhwan Kim, Younggyun Hahm, Zhong He, Tony Kyungil Lee, Enrico Santus, Francis Bond, and Seung-Hoon Na, editors, *Proceedings of the 29th International Conference on Computational Linguistics*, pages 3563–3569, Gyeongju, Republic of Korea, October 2022. International Committee on Computational Linguistics. URL <https://aclanthology.org/2022.coling-1.314>.

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